

V4 parameter estimation

Seven models · SPY, AAPL, MSFT, AMZN · four windows · 1.5-year fixed calibration

What this report is

- The single t_0 parameter vector used in this study, summarized by regime. One table per model. $n = 1$ so SD is —.
- No LSM, no option prices, and no simulated paths. Only calibrate_ticker with the 18-month lookback.
- The companion notebook is the computational source of truth. This PDF is written from the same payload.

How estimates are formed

- At the first session t_0 of each 1-year evaluation window, parameters use only returns and listed quotes with `trading_date ≤ t0`. Those parameters are held for every Monday contract in the window. LSM prices each contract with that single calibration row. Option quotes are as-of the grid timestamp (same session, else the prior session).
- Calibration window: 1.5-year ending at the first session of each 1-year evaluation window. Rolling: none (one estimate, held for the whole window). Parameters estimated at t_0 are used only after t_0 .
- Cells are Mean, SD, Min, and Max. With $n = 1$, Mean = Min = Max and SD is —.

Tables

- P1 GBM. P2 Modified GBM. P3 GARCH. P4 Heston. P5 Merton. P6 GARCH-Merton. P7 Heston-Merton.
- R1 Crisis, R2 Normal, R3 Late-cycle, R4 COVID. Columns are SPY, AAPL, MSFT, AMZN.

Table P1 · GBM · none parameter estimates

1.5-year lookback · rolling=none · Mean, SD, Min, Max within each regime

gime / para	SPY				AAPL				MSFT				AMZN			
gime / para	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
μ (P)	-0.0912	—	-0.0912	-0.0912	0.4097	—	0.4097	0.4097	-0.1082	—	-0.1082	-0.1082	0.4477	—	0.4477	0.4477
σ	0.1852	—	0.1852	0.1852	0.4044	—	0.4044	0.4044	0.2730	—	0.2730	0.2730	0.5045	—	0.5045	0.5045
R2 Normal																
μ (P)	0.1945	—	0.1945	0.1945	-0.0234	—	-0.0234	-0.0234	0.1597	—	0.1597	0.1597	0.3665	—	0.3665	0.3665
σ	0.1147	—	0.1147	0.1147	0.2892	—	0.2892	0.2892	0.2339	—	0.2339	0.2339	0.2668	—	0.2668	0.2668
R3 Late-cycle																
μ (P)	0.1419	—	0.1419	0.1419	0.3211	—	0.3211	0.3211	0.3944	—	0.3944	0.3944	0.5438	—	0.5438	0.5438
σ	0.1100	—	0.1100	0.1100	0.2073	—	0.2073	0.2073	0.2007	—	0.2007	0.2007	0.2412	—	0.2412	0.2412
R4 COVID																
μ (P)	0.0514	—	0.0514	0.0514	0.1183	—	0.1183	0.1183	0.2690	—	0.2690	0.2690	0.1177	—	0.1177	0.1177
σ	0.1544	—	0.1544	0.1544	0.2924	—	0.2924	0.2924	0.2562	—	0.2562	0.2562	0.3257	—	0.3257	0.3257

Notes: This table reports Mean, SD, Min, and Max of the none GBM estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ (P), σ . Each company×regime cell is one estimate: 18-month lookback ending on the first session of the window, held for the year. n is 1. SD is undefined with a single update (shown as —). This is parameter estimation only; it is not an LSM pricing table.

Table P2 · Modified GBM · none parameter estimates

1.5-year lookback · rolling=none · Mean, SD, Min, Max within each regime

Regime / parameter	SPY				AAPL				MSFT				AMZN			
Regime / parameter	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
P(U U)	0.4602	—	0.4602	0.4602	0.5071	—	0.5071	0.5071	0.4918	—	0.4918	0.4918	0.4335	—	0.4335	0.4335
P(D D)	0.3961	—	0.3961	0.3961	0.3899	—	0.3899	0.3899	0.5079	—	0.5079	0.5079	0.4395	—	0.4395	0.4395
P(U D)	0.6039	—	0.6039	0.6039	0.6101	—	0.6101	0.6101	0.4921	—	0.4921	0.4921	0.5605	—	0.5605	0.5605
P(D U)	0.5398	—	0.5398	0.5398	0.4929	—	0.4929	0.4929	0.5082	—	0.5082	0.5082	0.5665	—	0.5665	0.5665
μ_U	0.0077	—	0.0077	0.0077	0.0190	—	0.0190	0.0190	0.0127	—	0.0127	0.0127	0.0230	—	0.0230	0.0230
σ_U	0.0073	—	0.0073	0.0073	0.0148	—	0.0148	0.0148	0.0118	—	0.0118	0.0118	0.0293	—	0.0293	0.0293
μ_D	0.0095	—	0.0095	0.0095	0.0200	—	0.0200	0.0200	0.0131	—	0.0131	0.0131	0.0191	—	0.0191	0.0191
σ_D	0.0085	—	0.0085	0.0085	0.0185	—	0.0185	0.0185	0.0113	—	0.0113	0.0113	0.0171	—	0.0171	0.0171
R2 Normal																
P(U U)	0.5678	—	0.5678	0.5678	0.4572	—	0.4572	0.4572	0.4921	—	0.4921	0.4921	0.4974	—	0.4974	0.4974
P(D D)	0.4488	—	0.4488	0.4488	0.4738	—	0.4738	0.4738	0.4724	—	0.4724	0.4724	0.4947	—	0.4947	0.4947
P(U D)	0.5512	—	0.5512	0.5512	0.5262	—	0.5262	0.5262	0.5276	—	0.5276	0.5276	0.5053	—	0.5053	0.5053
P(D U)	0.4322	—	0.4322	0.4322	0.5428	—	0.5428	0.5428	0.5079	—	0.5079	0.5079	0.5026	—	0.5026	0.5026
μ_U	0.0055	—	0.0055	0.0055	0.0132	—	0.0132	0.0132	0.0107	—	0.0107	0.0107	0.0139	—	0.0139	0.0139
σ_U	0.0047	—	0.0047	0.0047	0.0110	—	0.0110	0.0110	0.0101	—	0.0101	0.0101	0.0125	—	0.0125	0.0125
μ_D	0.0054	—	0.0054	0.0054	0.0131	—	0.0131	0.0131	0.0099	—	0.0099	0.0099	0.0111	—	0.0111	0.0111
σ_D	0.0049	—	0.0049	0.0049	0.0142	—	0.0142	0.0142	0.0114	—	0.0114	0.0114	0.0100	—	0.0100	0.0100
R3 Late-cy																
P(U U)	0.5532	—	0.5532	0.5532	0.5371	—	0.5371	0.5371	0.5233	—	0.5233	0.5233	0.6149	—	0.6149	0.6149
P(D D)	0.4018	—	0.4018	0.4018	0.4688	—	0.4688	0.4688	0.3531	—	0.3531	0.3531	0.4554	—	0.4554	0.4554
P(U D)	0.5982	—	0.5982	0.5982	0.5312	—	0.5312	0.5312	0.6469	—	0.6469	0.6469	0.5446	—	0.5446	0.5446
P(D U)	0.4468	—	0.4468	0.4468	0.4629	—	0.4629	0.4629	0.4767	—	0.4767	0.4767	0.3851	—	0.3851	0.3851
μ_U	0.0045	—	0.0045	0.0045	0.0100	—	0.0100	0.0100	0.0092	—	0.0092	0.0092	0.0110	—	0.0110	0.0110
σ_U	0.0042	—	0.0042	0.0042	0.0095	—	0.0095	0.0095	0.0090	—	0.0090	0.0090	0.0112	—	0.0112	0.0112
μ_D	0.0047	—	0.0047	0.0047	0.0088	—	0.0088	0.0088	0.0087	—	0.0087	0.0087	0.0104	—	0.0104	0.0104
σ_D	0.0063	—	0.0063	0.0063	0.0086	—	0.0086	0.0086	0.0093	—	0.0093	0.0093	0.0106	—	0.0106	0.0106
R4 COVID																
P(U U)	0.5585	—	0.5585	0.5585	0.5515	—	0.5515	0.5515	0.5186	—	0.5186	0.5186	0.5738	—	0.5738	0.5738
P(D D)	0.4856	—	0.4856	0.4856	0.4829	—	0.4829	0.4829	0.3673	—	0.3673	0.3673	0.4673	—	0.4673	0.4673
P(U D)	0.5144	—	0.5144	0.5144	0.5171	—	0.5171	0.5171	0.6327	—	0.6327	0.6327	0.5327	—	0.5327	0.5327
P(D U)	0.4415	—	0.4415	0.4415	0.4485	—	0.4485	0.4485	0.4814	—	0.4814	0.4814	0.4262	—	0.4262	0.4262
μ_U	0.0065	—	0.0065	0.0065	0.0125	—	0.0125	0.0125	0.0112	—	0.0112	0.0112	0.0134	—	0.0134	0.0134
σ_U	0.0061	—	0.0061	0.0061	0.0117	—	0.0117	0.0117	0.0107	—	0.0107	0.0107	0.0135	—	0.0135	0.0135
μ_D	0.0072	—	0.0072	0.0072	0.0135	—	0.0135	0.0135	0.0124	—	0.0124	0.0124	0.0157	—	0.0157	0.0157
σ_D	0.0078	—	0.0078	0.0078	0.0146	—	0.0146	0.0146	0.0118	—	0.0118	0.0118	0.0159	—	0.0159	0.0159

Notes: This table reports Mean, SD, Min, and Max of the none Modified GBM estimates for SPY, AAPL, MSFT, AMZN. Parameters are P(U|U), P(D|D), P(U|D), P(D|U), μ_U, σ_U, μ_D, σ_D. Each company×regime cell is one estimate: 18-month lookback ending on the first session of the window, held for the year. n is 1. SD is undefined with a single update (shown as —). This is parameter estimation only; it is not an LSM pricing table.

Table P3 · GARCH · none parameter estimates

1.5-year lookback · rolling=none · Mean, SD, Min, Max within each regime

Regime / para	SPY				AAPL				MSFT				AMZN			
Regime / para	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
λ	-0.0135	—	-0.0135	-0.0135	0.1128	—	0.1128	0.1128	-0.0340	—	-0.0340	-0.0340	0.0778	—	0.0778	0.0778
ω	2.700e-06	—	2.700e-06	2.700e-06	4.945e-05	—	4.945e-05	4.945e-05	1.196e-05	—	1.196e-05	1.196e-05	9.008e-04	—	9.008e-04	9.008e-04
α	0.0553	—	0.0553	0.0553	0.0991	—	0.0991	0.0991	0.1027	—	0.1027	0.1027	0.1091	—	0.1091	0.1091
β	0.9276	—	0.9276	0.9276	0.8269	—	0.8269	0.8269	0.8667	—	0.8667	0.8667	1.568e-15	—	1.568e-15	1.568e-15
σ_0	0.0140	—	0.0140	0.0140	0.0261	—	0.0261	0.0261	0.0239	—	0.0239	0.0239	0.0312	—	0.0312	0.0312
persist P	0.9829	—	0.9829	0.9829	0.9260	—	0.9260	0.9260	0.9694	—	0.9694	0.9694	0.1091	—	0.1091	0.1091
persist Q	0.9829	—	0.9829	0.9829	0.9273	—	0.9273	0.9273	0.9695	—	0.9695	0.9695	0.1097	—	0.1097	0.1097
R2 Normal																
λ	0.1383	—	0.1383	0.1383	0.0364	—	0.0364	0.0364	0.0695	—	0.0695	0.0695	0.1122	—	0.1122	0.1122
ω	1.292e-05	—	1.292e-05	1.292e-05	6.448e-06	—	6.448e-06	6.448e-06	1.489e-04	—	1.489e-04	1.489e-04	4.240e-05	—	4.240e-05	4.240e-05
α	0.1340	—	0.1340	0.1340	0.0381	—	0.0381	0.0381	0.1365	—	0.1365	0.1365	0.1213	—	0.1213	0.1213
β	0.6231	—	0.6231	0.6231	0.9441	—	0.9441	0.9441	0.2046	—	0.2046	0.2046	0.7372	—	0.7372	0.7372
σ_0	0.0061	—	0.0061	0.0061	0.0152	—	0.0152	0.0152	0.0137	—	0.0137	0.0137	0.0154	—	0.0154	0.0154
persist P	0.7571	—	0.7571	0.7571	0.9822	—	0.9822	0.9822	0.3412	—	0.3412	0.3412	0.8585	—	0.8585	0.8585
persist Q	0.7597	—	0.7597	0.7597	0.9823	—	0.9823	0.9823	0.3418	—	0.3418	0.3418	0.8600	—	0.8600	0.8600
R3 Late-cy																
λ	0.1689	—	0.1689	0.1689	0.1199	—	0.1199	0.1199	0.1720	—	0.1720	0.1720	0.1899	—	0.1899	0.1899
ω	2.976e-06	—	2.976e-06	2.976e-06	4.363e-05	—	4.363e-05	4.363e-05	4.227e-05	—	4.227e-05	4.227e-05	6.055e-05	—	6.055e-05	6.055e-05
α	0.2165	—	0.2165	0.2165	0.1233	—	0.1233	0.1233	0.2008	—	0.2008	0.2008	0.1293	—	0.1293	0.1293
β	0.7216	—	0.7216	0.7216	0.6206	—	0.6206	0.6206	0.5159	—	0.5159	0.5159	0.6160	—	0.6160	0.6160
σ_0	0.0044	—	0.0044	0.0044	0.0124	—	0.0124	0.0124	0.0096	—	0.0096	0.0096	0.0143	—	0.0143	0.0143
persist P	0.9381	—	0.9381	0.9381	0.7440	—	0.7440	0.7440	0.7167	—	0.7167	0.7167	0.7453	—	0.7453	0.7453
persist Q	0.9443	—	0.9443	0.9443	0.7458	—	0.7458	0.7458	0.7226	—	0.7226	0.7226	0.7500	—	0.7500	0.7500
R4 COVID																
λ	0.1093	—	0.1093	0.1093	0.1151	—	0.1151	0.1151	0.1191	—	0.1191	0.1191	0.1185	—	0.1185	0.1185
ω	4.722e-06	—	4.722e-06	4.722e-06	1.901e-05	—	1.901e-05	1.901e-05	2.029e-05	—	2.029e-05	2.029e-05	1.594e-05	—	1.594e-05	1.594e-05
α	0.1677	—	0.1677	0.1677	0.1625	—	0.1625	0.1625	0.1842	—	0.1842	0.1842	0.2019	—	0.2019	0.2019
β	0.7890	—	0.7890	0.7890	0.7999	—	0.7999	0.7999	0.7450	—	0.7450	0.7450	0.7701	—	0.7701	0.7701
σ_0	0.0106	—	0.0106	0.0106	0.0186	—	0.0186	0.0186	0.0143	—	0.0143	0.0143	0.0142	—	0.0142	0.0142
persist P	0.9567	—	0.9567	0.9567	0.9624	—	0.9624	0.9624	0.9292	—	0.9292	0.9292	0.9720	—	0.9720	0.9720
persist Q	0.9587	—	0.9587	0.9587	0.9645	—	0.9645	0.9645	0.9318	—	0.9318	0.9318	0.9748	—	0.9748	0.9748

Notes: This table reports Mean, SD, Min, and Max of the none GARCH estimates for SPY, AAPL, MSFT, AMZN. Parameters are λ , ω , α , β , σ_0 , persist P, persist Q. Each company×regime cell is one estimate: 18-month lookback ending on the first session of the window, held for the year. n is 1. SD is undefined with a single update (shown as —). This is parameter estimation only; it is not an LSM pricing table.

Table P4 · Heston · none parameter estimates

1.5-year lookback · rolling=none · Mean, SD, Min, Max within each regime

Regime / parameter	SPY				AAPL				MSFT				AMZN			
Regime / parameter	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
μ (P)	-0.0912	—	-0.0912	-0.0912	0.4097	—	0.4097	0.4097	-0.1082	—	-0.1082	-0.1082	0.4477	—	0.4477	0.4477
κ (P)	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000
θ (P)	0.0343	—	0.0343	0.0343	0.1638	—	0.1638	0.1638	0.0744	—	0.0744	0.0744	0.2547	—	0.2547	0.2547
ξ	3.0000	—	3.0000	3.0000	3.0000	—	3.0000	3.0000	3.0000	—	3.0000	3.0000	3.0000	—	3.0000	3.0000
ρ	-0.1144	—	-0.1144	-0.1144	-0.0762	—	-0.0762	-0.0762	0.0602	—	0.0602	0.0602	0.3536	—	0.3536	0.3536
v_0 (P)	0.0497	—	0.0497	0.0497	0.1535	—	0.1535	0.1535	0.1522	—	0.1522	0.1522	0.3570	—	0.3570	0.3570
η_v (premium)	2.2564	—	2.2564	2.2564	1.8025	—	1.8025	1.8025	-0.1928	—	-0.1928	-0.1928	-2.5579	—	-2.5579	-2.5579
κ^* (Q)	17.7436	—	17.7436	17.7436	18.1975	—	18.1975	18.1975	20.1928	—	20.1928	20.1928	22.5579	—	22.5579	22.5579
θ^* (Q)	0.0386	—	0.0386	0.0386	0.1800	—	0.1800	0.1800	0.0737	—	0.0737	0.0737	0.2258	—	0.2258	0.2258
$v_0^{\wedge}Q$	0.0234	—	0.0234	0.0234	0.1424	—	0.1424	0.1424	0.1186	—	0.1186	0.1186	0.1615	—	0.1615	0.1615
n quotes	23.0000	—	23.0000	23.0000	12.0000	—	12.0000	12.0000	10.0000	—	10.0000	10.0000	8.0000	—	8.0000	8.0000
R2 Normal																
μ (P)	0.1945	—	0.1945	0.1945	-0.0234	—	-0.0234	-0.0234	0.1597	—	0.1597	0.1597	0.3665	—	0.3665	0.3665
κ (P)	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000
θ (P)	0.0133	—	0.0133	0.0133	0.0834	—	0.0834	0.0834	0.0547	—	0.0547	0.0547	0.0715	—	0.0715	0.0715
ξ	3.0000	—	3.0000	3.0000	3.0000	—	3.0000	3.0000	3.0000	—	3.0000	3.0000	3.0000	—	3.0000	3.0000
ρ	0.0165	—	0.0165	0.0165	-0.1986	—	-0.1986	-0.1986	-0.1805	—	-0.1805	-0.1805	0.2440	—	0.2440	0.2440
v_0 (P)	0.0094	—	0.0094	0.0094	0.0425	—	0.0425	0.0425	0.0308	—	0.0308	0.0308	0.0309	—	0.0309	0.0309
η_v (premium)	-40.0000	—	-40.0000	-40.0000	13.2632	—	13.2632	13.2632	6.8400	—	6.8400	6.8400	19.2681	—	19.2681	19.2681
κ^* (Q)	60.0000	—	60.0000	60.0000	6.7368	—	6.7368	6.7368	13.1600	—	13.1600	13.1600	0.7319	—	0.7319	0.7319
θ^* (Q)	0.0044	—	0.0044	0.0044	0.2477	—	0.2477	0.2477	0.0831	—	0.0831	0.0831	1.9540	—	1.9540	1.9540
$v_0^{\wedge}Q$	0.0336	—	0.0336	0.0336	0.0533	—	0.0533	0.0533	0.0543	—	0.0543	0.0543	0.0481	—	0.0481	0.0481
n quotes	22.0000	—	22.0000	22.0000	24.0000	—	24.0000	24.0000	20.0000	—	20.0000	20.0000	24.0000	—	24.0000	24.0000
R3 Late-cy																
μ (P)	0.1419	—	0.1419	0.1419	0.3211	—	0.3211	0.3211	0.3944	—	0.3944	0.3944	0.5438	—	0.5438	0.5438
κ (P)	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000
θ (P)	0.0121	—	0.0121	0.0121	0.0433	—	0.0433	0.0433	0.0408	—	0.0408	0.0408	0.0592	—	0.0592	0.0592
ξ	3.0000	—	3.0000	3.0000	3.0000	—	3.0000	3.0000	3.0000	—	3.0000	3.0000	3.0000	—	3.0000	3.0000
ρ	-0.3910	—	-0.3910	-0.3910	0.1557	—	0.1557	0.1557	0.1792	—	0.1792	0.1792	0.2105	—	0.2105	0.2105
v_0 (P)	0.0032	—	0.0032	0.0032	0.0465	—	0.0465	0.0465	0.0281	—	0.0281	0.0281	0.0528	—	0.0528	0.0528
η_v (premium)	-29.9841	—	-29.9841	-29.9841	17.8333	—	17.8333	17.8333	9.2254	—	9.2254	9.2254	19.3685	—	19.3685	19.3685
κ^* (Q)	49.9841	—	49.9841	49.9841	2.1667	—	2.1667	2.1667	10.7746	—	10.7746	10.7746	0.6315	—	0.6315	0.6315
θ^* (Q)	0.0049	—	0.0049	0.0049	0.3992	—	0.3992	0.3992	0.0757	—	0.0757	0.0757	1.8756	—	1.8756	1.8756
$v_0^{\wedge}Q$	0.0232	—	0.0232	0.0232	0.0132	—	0.0132	0.0132	0.0111	—	0.0111	0.0111	0.0356	—	0.0356	0.0356
n quotes	24.0000	—	24.0000	24.0000	24.0000	—	24.0000	24.0000	22.0000	—	22.0000	22.0000	24.0000	—	24.0000	24.0000
R4 COVID																
μ (P)	0.0514	—	0.0514	0.0514	0.1183	—	0.1183	0.1183	0.2690	—	0.2690	0.2690	0.1177	—	0.1177	0.1177
κ (P)	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000
θ (P)	0.0238	—	0.0238	0.0238	0.0853	—	0.0853	0.0853	0.0657	—	0.0657	0.0657	0.1059	—	0.1059	0.1059
ξ	3.0000	—	3.0000	3.0000	3.0000	—	3.0000	3.0000	3.0000	—	3.0000	3.0000	3.0000	—	3.0000	3.0000
ρ	-0.1561	—	-0.1561	-0.1561	-0.1852	—	-0.1852	-0.1852	0.0359	—	0.0359	0.0359	0.0188	—	0.0188	0.0188
v_0 (P)	0.0510	—	0.0510	0.0510	0.1299	—	0.1299	0.1299	0.0789	—	0.0789	0.0789	0.0681	—	0.0681	0.0681
η_v (premium)	-33.6565	—	-33.6565	-33.6565	7.9082	—	7.9082	7.9082	2.8034	—	2.8034	2.8034	1.7983	—	1.7983	1.7983
κ^* (Q)	53.6565	—	53.6565	53.6565	12.0918	—	12.0918	12.0918	17.1966	—	17.1966	17.1966	18.2017	—	18.2017	18.2017
θ^* (Q)	0.0089	—	0.0089	0.0089	0.1412	—	0.1412	0.1412	0.0765	—	0.0765	0.0765	0.1163	—	0.1163	0.1163
$v_0^{\wedge}Q$	0.0422	—	0.0422	0.0422	0.0638	—	0.0638	0.0638	0.0373	—	0.0373	0.0373	0.0106	—	0.0106	0.0106
n quotes	24.0000	—	24.0000	24.0000	24.0000	—	24.0000	24.0000	23.0000	—	23.0000	23.0000	24.0000	—	24.0000	24.0000

Notes: This table reports Mean, SD, Min, and Max of the none Heston estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ (P), κ (P), θ (P), ξ , ρ , v_0 (P), η_v (premium), κ^* (Q), θ^* (Q), $v_0^{\wedge}Q$, n quotes. Each company×regime cell is one estimate: 18-month lookback ending on the first session of the window, held for the year. n is 1. SD is undefined with a single update (shown as —). This is parameter estimation only; it is not an LSM pricing table.

Table P5 · Merton · none parameter estimates

1.5-year lookback · rolling=none · Mean, SD, Min, Max within each regime

Regime / parameter	SPY	SD	Min	Max	AAPL	SD	Min	Max	MSFT	SD	Min	Max	AMZN	SD	Min	Max
R1 Crisis	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
μ (P)	-0.1447	—	-0.1447	-0.1447	0.4746	—	0.4746	0.4746	-0.0396	—	-0.0396	-0.0396	0.0224	—	0.0224	0.0224
σ	0.1725	—	0.1725	0.1725	0.3680	—	0.3680	0.3680	0.2379	—	0.2379	0.2379	0.3758	—	0.3758	0.3758
$\lambda^* = \lambda$	2.6667	—	2.6667	2.6667	2.0000	—	2.0000	2.0000	2.6667	—	2.6667	2.6667	4.0000	—	4.0000	4.0000
μ_J (P)	0.0195	—	0.0195	0.0195	-0.0306	—	-0.0306	-0.0306	-0.0259	—	-0.0259	-0.0259	0.1064	—	0.1064	0.1064
σ_J	0.0364	—	0.0364	0.0364	0.1145	—	0.1145	0.1145	0.0778	—	0.0778	0.0778	0.1304	—	0.1304	0.1304
κ (P)	0.0203	—	0.0203	0.0203	-0.0238	—	-0.0238	-0.0238	-0.0226	—	-0.0226	-0.0226	0.1218	—	0.1218	0.1218
μ_J^* (Q)	-0.0158	—	-0.0158	-0.0158	-0.0587	—	-0.0587	-0.0587	-0.1056	—	-0.1056	-0.1056	-0.0999	—	-0.0999	-0.0999
κ^Q	-0.0151	—	-0.0151	-0.0151	-0.0508	—	-0.0508	-0.0508	-0.0975	—	-0.0975	-0.0975	-0.0873	—	-0.0873	-0.0873
n quotes	23.0000	—	23.0000	23.0000	12.0000	—	12.0000	12.0000	10.0000	—	10.0000	10.0000	8.0000	—	8.0000	8.0000
R2 Normal	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
μ (P)	0.2135	—	0.2135	0.2135	0.1389	—	0.1389	0.1389	0.2188	—	0.2188	0.2188	0.2677	—	0.2677	0.2677
σ	0.1041	—	0.1041	0.1041	0.2411	—	0.2411	0.2411	0.1799	—	0.1799	0.1799	0.2176	—	0.2176	0.2176
$\lambda^* = \lambda$	3.3245	—	3.3245	3.3245	3.3245	—	3.3245	3.3245	3.3245	—	3.3245	3.3245	3.9894	—	3.9894	3.9894
μ_J (P)	-0.0049	—	-0.0049	-0.0049	-0.0483	—	-0.0483	-0.0483	-0.0169	—	-0.0169	-0.0169	0.0258	—	0.0258	0.0258
σ_J	0.0259	—	0.0259	0.0259	0.0731	—	0.0731	0.0731	0.0802	—	0.0802	0.0802	0.0728	—	0.0728	0.0728
κ (P)	-0.0045	—	-0.0045	-0.0045	-0.0446	—	-0.0446	-0.0446	-0.0136	—	-0.0136	-0.0136	0.0289	—	0.0289	0.0289
μ_J^* (Q)	-0.0115	—	-0.0115	-0.0115	-0.0668	—	-0.0668	-0.0668	0.0092	—	0.0092	0.0092	0.0300	—	0.0300	0.0300
κ^Q	-0.0111	—	-0.0111	-0.0111	-0.0622	—	-0.0622	-0.0622	0.0125	—	0.0125	0.0125	0.0331	—	0.0331	0.0331
n quotes	22.0000	—	22.0000	22.0000	24.0000	—	24.0000	24.0000	20.0000	—	20.0000	20.0000	24.0000	—	24.0000	24.0000
R3 Late-cy	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
μ (P)	0.2578	—	0.2578	0.2578	0.2281	—	0.2281	0.2281	0.4313	—	0.4313	0.4313	0.5329	—	0.5329	0.5329
σ	0.0866	—	0.0866	0.0866	0.1723	—	0.1723	0.1723	0.1625	—	0.1625	0.1625	0.1942	—	0.1942	0.1942
$\lambda^* = \lambda$	5.3333	—	5.3333	5.3333	6.0000	—	6.0000	6.0000	4.0000	—	4.0000	4.0000	2.0000	—	2.0000	2.0000
μ_J (P)	-0.0207	—	-0.0207	-0.0207	0.0164	—	0.0164	0.0164	-0.0075	—	-0.0075	-0.0075	0.0076	—	0.0076	0.0076
σ_J	0.0208	—	0.0208	0.0208	0.0440	—	0.0440	0.0440	0.0585	—	0.0585	0.0585	0.1010	—	0.1010	0.1010
κ (P)	-0.0203	—	-0.0203	-0.0203	0.0175	—	0.0175	0.0175	-0.0058	—	-0.0058	-0.0058	0.0128	—	0.0128	0.0128
μ_J^* (Q)	-0.0089	—	-0.0089	-0.0089	-0.0547	—	-0.0547	-0.0547	-0.0511	—	-0.0511	-0.0511	0.0288	—	0.0288	0.0288
κ^Q	-0.0086	—	-0.0086	-0.0086	-0.0523	—	-0.0523	-0.0523	-0.0482	—	-0.0482	-0.0482	0.0344	—	0.0344	0.0344
n quotes	24.0000	—	24.0000	24.0000	24.0000	—	24.0000	24.0000	22.0000	—	22.0000	22.0000	24.0000	—	24.0000	24.0000
R4 COVID	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
μ (P)	0.1032	—	0.1032	0.1032	0.1487	—	0.1487	0.1487	0.2152	—	0.2152	0.2152	0.1095	—	0.1095	0.1095
σ	0.1319	—	0.1319	0.1319	0.2455	—	0.2455	0.2455	0.2236	—	0.2236	0.2236	0.2835	—	0.2835	0.2835
$\lambda^* = \lambda$	4.6667	—	4.6667	4.6667	4.0000	—	4.0000	4.0000	3.3333	—	3.3333	3.3333	4.0000	—	4.0000	4.0000
μ_J (P)	-0.0107	—	-0.0107	-0.0107	-0.0070	—	-0.0070	-0.0070	0.0170	—	0.0170	0.0170	0.0025	—	0.0025	0.0025
σ_J	0.0356	—	0.0356	0.0356	0.0791	—	0.0791	0.0791	0.0664	—	0.0664	0.0664	0.0802	—	0.0802	0.0802
κ (P)	-0.0100	—	-0.0100	-0.0100	-0.0039	—	-0.0039	-0.0039	0.0194	—	0.0194	0.0194	0.0057	—	0.0057	0.0057
μ_J^* (Q)	-0.0139	—	-0.0139	-0.0139	-0.0550	—	-0.0550	-0.0550	-0.0373	—	-0.0373	-0.0373	-0.0499	—	-0.0499	-0.0499
κ^Q	-0.0132	—	-0.0132	-0.0132	-0.0506	—	-0.0506	-0.0506	-0.0345	—	-0.0345	-0.0345	-0.0456	—	-0.0456	-0.0456
n quotes	24.0000	—	24.0000	24.0000	24.0000	—	24.0000	24.0000	23.0000	—	23.0000	23.0000	24.0000	—	24.0000	24.0000

Notes: This table reports Mean, SD, Min, and Max of the none Merton estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ (P), σ , $\lambda^* = \lambda$, μ_J (P), σ_J , κ (P), μ_J^* (Q), κ^Q , n quotes. Each company×regime cell is one estimate: 18-month lookback ending on the first session of the window, held for the year. n is 1. SD is undefined with a single update (shown as —). This is parameter estimation only; it is not an LSM pricing table.

Table P6 · GARCH-Merton · none parameter estimates

1.5-year lookback · rolling=none · Mean, SD, Min, Max within each regime

Regime / parameter	SPY				AAPL				MSFT				AMZN			
Regime / parameter	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
μ	-0.0305	—	-0.0305	-0.0305	0.6989	—	0.6989	0.6989	-0.0817	—	-0.0817	-0.0817	0.5326	—	0.5326	0.5326
ω	2.695e-06	—	2.695e-06	2.695e-06	4.710e-05	—	4.710e-05	4.710e-05	1.204e-05	—	1.204e-05	1.204e-05	9.001e-04	—	9.001e-04	9.001e-04
α	0.0554	—	0.0554	0.0554	0.0987	—	0.0987	0.0987	0.1011	—	0.1011	0.1011	0.1101	—	0.1101	0.1101
β	0.9275	—	0.9275	0.9275	0.8310	—	0.8310	0.8310	0.8678	—	0.8678	0.8678	3.061e-14	—	3.061e-14	3.061e-14
σ_0	0.0140	—	0.0140	0.0140	0.0262	—	0.0262	0.0262	0.0238	—	0.0238	0.0238	0.0312	—	0.0312	0.0312
λ	0.6667	—	0.6667	0.6667	2.0000	—	2.0000	2.0000	2.0000	—	2.0000	2.0000	2.6667	—	2.6667	2.6667
μ_J	-0.0350	—	-0.0350	-0.0350	-0.0851	—	-0.0851	-0.0851	-0.0138	—	-0.0138	-0.0138	0.1666	—	0.1666	0.1666
σ_J	0.0000	—	0.0000	0.0000	0.0251	—	0.0251	0.0251	0.0906	—	0.0906	0.0906	0.0723	—	0.0723	0.0723
κ	-0.0344	—	-0.0344	-0.0344	-0.0813	—	-0.0813	-0.0813	-0.0096	—	-0.0096	-0.0096	0.1844	—	0.1844	0.1844
R2 Normal																
μ	0.2315	—	0.2315	0.2315	0.1421	—	0.1421	0.1421	0.2948	—	0.2948	0.2948	0.4583	—	0.4583	0.4583
ω	1.379e-05	—	1.379e-05	1.379e-05	6.339e-06	—	6.339e-06	6.339e-06	1.340e-04	—	1.340e-04	1.340e-04	4.158e-05	—	4.158e-05	4.158e-05
α	0.1358	—	0.1358	0.1358	0.0382	—	0.0382	0.0382	0.1734	—	0.1734	0.1734	0.1263	—	0.1263	0.1263
β	0.6047	—	0.6047	0.6047	0.9443	—	0.9443	0.9443	0.2481	—	0.2481	0.2481	0.7358	—	0.7358	0.7358
σ_0	0.0061	—	0.0061	0.0061	0.0151	—	0.0151	0.0151	0.0134	—	0.0134	0.0134	0.0154	—	0.0154	0.0154
λ	1.9947	—	1.9947	1.9947	3.9894	—	3.9894	3.9894	2.6596	—	2.6596	2.6596	3.3245	—	3.3245	3.3245
μ_J	-0.0087	—	-0.0087	-0.0087	-0.0319	—	-0.0319	-0.0319	-0.0098	—	-0.0098	-0.0098	0.0427	—	0.0427	0.0427
σ_J	0.0251	—	0.0251	0.0251	0.0767	—	0.0767	0.0767	0.0908	—	0.0908	0.0908	0.0670	—	0.0670	0.0670
κ	-0.0083	—	-0.0083	-0.0083	-0.0285	—	-0.0285	-0.0285	-0.0057	—	-0.0057	-0.0057	0.0460	—	0.0460	0.0460
R3 Late-cy																
μ	0.2143	—	0.2143	0.2143	0.3426	—	0.3426	0.3426	0.4942	—	0.4942	0.4942	0.7010	—	0.7010	0.7010
ω	2.969e-06	—	2.969e-06	2.969e-06	4.492e-05	—	4.492e-05	4.492e-05	4.208e-05	—	4.208e-05	4.208e-05	5.971e-05	—	5.971e-05	5.971e-05
α	0.2082	—	0.2082	0.2082	0.1209	—	0.1209	0.1209	0.2060	—	0.2060	0.2060	0.1350	—	0.1350	0.1350
β	0.7275	—	0.7275	0.7275	0.6152	—	0.6152	0.6152	0.5131	—	0.5131	0.5131	0.6153	—	0.6153	0.6153
σ_0	0.0044	—	0.0044	0.0044	0.0125	—	0.0125	0.0125	0.0096	—	0.0096	0.0096	0.0143	—	0.0143	0.0143
λ	3.3333	—	3.3333	3.3333	3.3333	—	3.3333	3.3333	2.6667	—	2.6667	2.6667	1.3333	—	1.3333	1.3333
μ_J	-0.0241	—	-0.0241	-0.0241	-0.0029	—	-0.0029	-0.0029	0.0169	—	0.0169	0.0169	0.0406	—	0.0406	0.0406
σ_J	0.0107	—	0.0107	0.0107	0.0502	—	0.0502	0.0502	0.0588	—	0.0588	0.0588	0.1181	—	0.1181	0.1181
κ	-0.0238	—	-0.0238	-0.0238	-0.0017	—	-0.0017	-0.0017	0.0188	—	0.0188	0.0188	0.0488	—	0.0488	0.0488
R4 COVID																
μ	0.2136	—	0.2136	0.2136	0.5268	—	0.5268	0.5268	0.3807	—	0.3807	0.3807	0.5059	—	0.5059	0.5059
ω	4.718e-06	—	4.718e-06	4.718e-06	1.819e-05	—	1.819e-05	1.819e-05	2.013e-05	—	2.013e-05	2.013e-05	1.517e-05	—	1.517e-05	1.517e-05
α	0.1684	—	0.1684	0.1684	0.1773	—	0.1773	0.1773	0.1825	—	0.1825	0.1825	0.2090	—	0.2090	0.2090
β	0.7886	—	0.7886	0.7886	0.7917	—	0.7917	0.7917	0.7474	—	0.7474	0.7474	0.7677	—	0.7677	0.7677
σ_0	0.0107	—	0.0107	0.0107	0.0186	—	0.0186	0.0186	0.0143	—	0.0143	0.0143	0.0142	—	0.0142	0.0142
λ	3.3333	—	3.3333	3.3333	5.3333	—	5.3333	5.3333	2.6667	—	2.6667	2.6667	1.3333	—	1.3333	1.3333
μ_J	-0.0292	—	-0.0292	-0.0292	-0.0204	—	-0.0204	-0.0204	-0.0175	—	-0.0175	-0.0175	-0.0151	—	-0.0151	-0.0151
σ_J	0.0041	—	0.0041	0.0041	0.0666	—	0.0666	0.0666	0.0613	—	0.0613	0.0613	0.0684	—	0.0684	0.0684
κ	-0.0288	—	-0.0288	-0.0288	-0.0180	—	-0.0180	-0.0180	-0.0155	—	-0.0155	-0.0155	-0.0127	—	-0.0127	-0.0127

Notes: This table reports Mean, SD, Min, and Max of the none GARCH-Merton estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ , ω , α , β , σ_0 , λ , μ_J , σ_J , κ . Each company×regime cell is one estimate: 18-month lookback ending on the first session of the window, held for the year. n is 1. SD is undefined with a single update (shown as —). This is parameter estimation only; it is not an LSM pricing table.

1.5-year lookback · rolling=none · Mean, SD, Min, Max within each regime

Notes: This table reports Mean, SD, Min, and Max of the none Heston-Merton estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ (P), κ (P), θ (P), ξ , ρ , v_0 (P), $\lambda^* = \lambda$, μ_J (P), σ_J , κ_J (P), η_{-v} (premium), κ^* (Q), θ^* (Q), v_0^*Q , μ_J^* (Q), κ_J^*Q , n quotes. Each company $\times$ regime cell is one estimate: 18-month lookback ending on the first session of the window, held for the year. n is 1. SD is undefined with a single update (shown as —). This is parameter estimation only; it is not an LSM pricing table.